

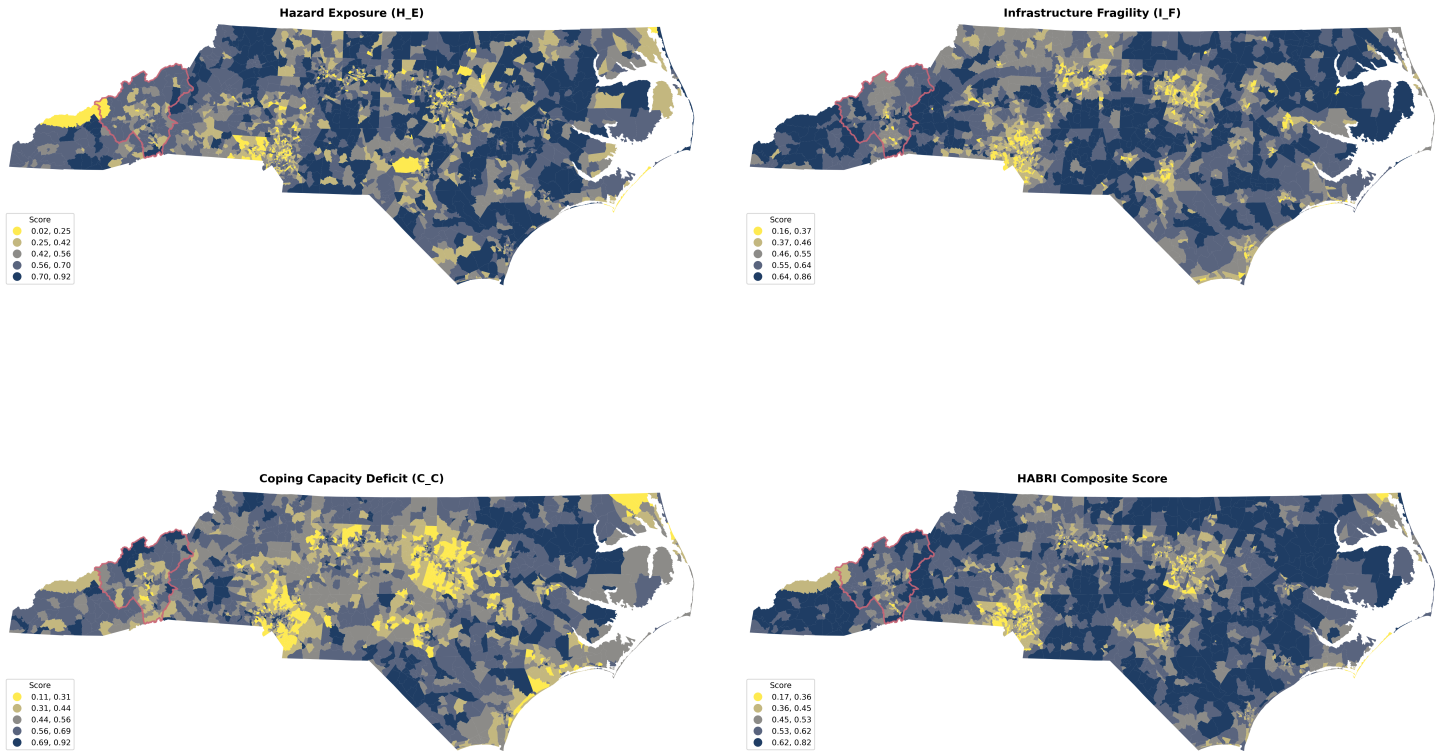
HABRI Variable Reference Sheet

Hazard-Adjusted Broadband Reliability Index — North Carolina, All 100 Counties (2,660 Census Tracts)

$$\text{HABRI} = 0.40 \times \text{Hazard Exposure (H_E)} + 0.35 \times \text{Infrastructure Fragility (I_F)} + 0.25 \times \text{Coping Capacity Deficit (C_C)}$$

Score range: 0 (lowest risk) to 1 (highest risk) | Higher scores indicate greater risk of broadband failure during a natural disaster

Hazard-Adjusted Broadband Reliability Index (HABRI) — North Carolina
All 2,660 Census Tracts | WNC counties outlined



Statewide maps of each sub-index and the final composite HABRI score. Darker shading = higher risk. Western NC validation counties outlined.

Hazard Exposure (H_E) — Weight: 0.40 — "How likely is a damaging natural hazard event?"			
Variable	Weight	Data Source	What It Measures and Why It Matters
Inland Flood Risk	0.40	FEMA National Risk Index v1.20 (federal hazard assessment)	Overall flood risk for each tract, combining likelihood with expected damage. Flooding is the most common NC hazard and directly damages broadband equipment in ground-level cabinets and underground conduit.
Hurricane Risk	0.35	FEMA National Risk Index v1.20 (federal hazard assessment)	Combined threat from hurricane winds, heavy rain, and coastal storm surge. Hurricanes cause the longest broadband outages because they knock out both the power grid and communications infrastructure simultaneously.
Landslide Risk	0.25	FEMA National Risk Index v1.20 (federal hazard assessment)	Susceptibility to rain-triggered slope failures in western Appalachian mountains. Landslides physically sever buried fiber-optic cables and block roads to remote cell tower sites.
Infrastructure Fragility (I_F) — Weight: 0.35 — "How resilient is the local broadband and access infrastructure?"			
Variable	Weight	Data Source	What It Measures and Why It Matters
Cellular Tower Density (fewer = higher risk)	0.30	Homeland Infrastructure Foundation (HIFLD), U.S. DHS (1,275 NC towers)	Cell towers per sq km. Areas with fewer towers have less redundancy — a single tower failure leaves a larger area without wireless service. Best publicly available proxy for wireless network resilience.
Broadband Latency (higher = higher risk)	0.30	Ookla Speedtest Open Data, Q3 2024 (~123,000 NC measurement points)	Delay in data transmission (milliseconds). High baseline latency signals aging or overloaded network equipment more prone to failure during disasters. Better indicator of infrastructure health than download speed alone.
Road & Utility Corridor Fragility (composite)	0.40	OpenStreetMap road network (648K intersections, 1.5M road segments)	Roads are the corridors along which fiber-optic cable, coaxial cable, and power lines are physically installed in rights-of-way. When a road is severed by flooding or landslide, it often destroys the broadband infrastructure buried alongside it and simultaneously prevents repair crews from reaching the damage. Measures both how critical each road is and how many alternative routes exist.
Coping Capacity Deficit (C_C) — Weight: 0.25 — "How well can residents cope when internet service is lost?"			
Variable	Weight	Data Source	What It Measures and Why It Matters
Households Without a Vehicle	0.20	U.S. Census Bureau, American Community Survey 2018–2022	Residents without a car cannot drive to a library, community center, or neighboring town with working internet during an outage. They are effectively stranded without connectivity.
Mobile-Only Internet Households	0.20	U.S. Census Bureau, American Community Survey 2018–2022	Households relying solely on cellular data (no home cable or fiber) have no backup if cell towers fail. When wireless goes down, they lose all internet access completely.
Disability Rate	0.20	U.S. Census Bureau, American Community Survey 2018–2022	People with disabilities depend more on internet-connected services (telehealth, medication management, assistive technology) and face greater physical barriers to reaching alternative access points.
Median Household Income (lower = risk)	0.20	U.S. Census Bureau, American Community Survey 2018–2022	Lower-income households have fewer resources to purchase backup connectivity (mobile hotspots, satellite service, generators) or absorb the economic cost of extended outages.
Poverty Rate	0.20	U.S. Census Bureau, American Community Survey 2018–2022	Poverty amplifies every other coping deficit. Households below the poverty line face the most constrained budgets for adaptation and experience the slowest recovery after any disaster.

Normalization: All variables are statistically standardized (z-score) then converted to a 0–1 probability scale where 0.5 = statewide average. For variables where higher raw values mean lower risk (tower density, road density, income), the scale is inverted so higher scores always mean higher risk. Missing values (48 tracts missing income) filled with statewide median. Spatial unit: 2020 Census tracts, NC State Plane coordinates (EPSG:2264). | Contact: Jesse R. Andrews, Texas Tech University (jesseand@ttu.edu)